Andrew Reed *reedas1* (*#94006807*)

CS121 Assignment 3 Milestone 2 Report

The group consists of one Software Engineering major, so provided metrics reflect the Algorithms and Data Structures Developer specifications.

Searching has been implemented with a preliminary form of result ranking (TF-IDF weighted by text importance), which influences the outputs shown for each of the queries below. All results have been inspected to verify that the searched query terms do appear in the page; pages with higher incidence are confirmed to rank higher to some degree, although text importance weighting remains to be fully balanced.

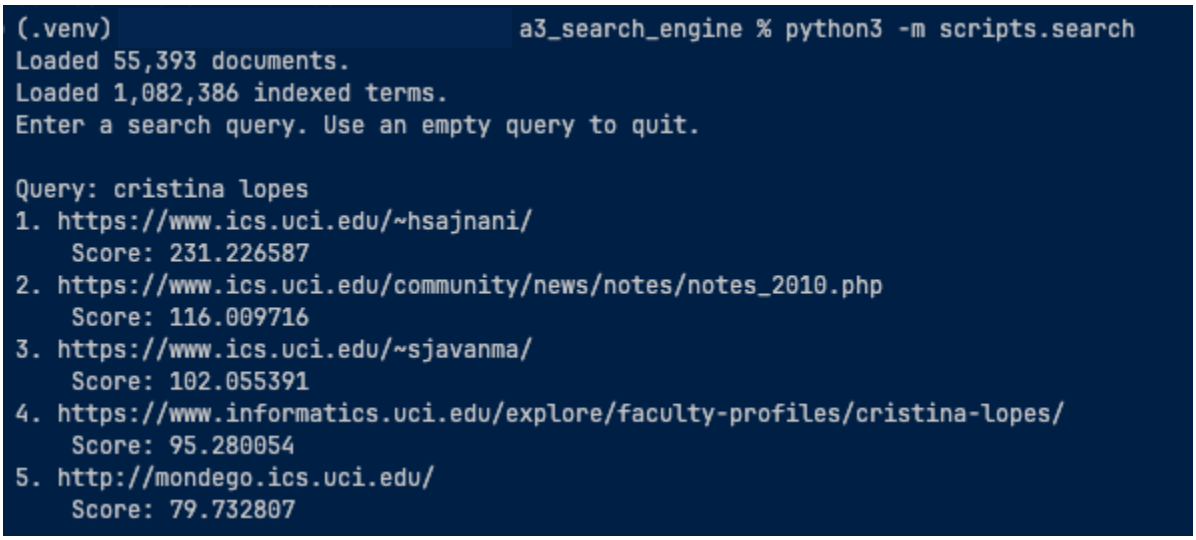
Query 1

Query 1 is **“cristina lopes”**.

Results

Rank	URL	Score
1	https://www.ics.uci.edu/~hsajnani/	231.226587
2	https://www.ics.uci.edu/community/news/notes/notes_2010.php	116.009716
3	https://www.ics.uci.edu/~sjavanma/	102.055391
4	https://www.informatics.uci.edu/explore/faculty-profiles/cristina-lopes/	95.280054
5	http://mondego.ics.uci.edu/	79.732807

Screenshot



Query 2

Query 2 is "**machine learning**".

Results

Rank	URL	Score
1	http://mondego.ics.uci.edu/datasets/maven-contents.txt	1129.465597
2	https://cml.ics.uci.edu/category/aiml/page/2/	533.433817
3	https://cml.ics.uci.edu/category/aiml/	487.634050
4	https://www.ics.uci.edu/faculty/index.php	423.289227
5	https://www.ics.uci.edu/faculty/	423.289227

Note: the rank 4 and 5 entries, despite being unique URLs, seem to have duplicate content. Disregarding the duplicate URL, the following can be considered an alternate rank 5 entry:

6	https://www.ics.uci.edu/~pazzani/Publications/OldPublications.html	360.668829

Screenshot

```
Query: machine learning
1. http://mondego.ics.uci.edu/datasets/maven-contents.txt
    Score: 1129.465597
2. https://cml.ics.uci.edu/category/aiml/page/2/
    Score: 533.433817
3. https://cml.ics.uci.edu/category/aiml/
    Score: 487.634050
4. https://www.ics.uci.edu/faculty/index.php
    Score: 423.289227
5. https://www.ics.uci.edu/faculty/
    Score: 423.289227
```

Query 3

Query 3 is “ACM”.

Results

Rank	URL	Score
1	https://www.ics.uci.edu/~eppstein/pubs/pubs.ff	421.727496
2	https://www.ics.uci.edu/~eppstein/pubs/all.html	389.077625
3	https://www.ics.uci.edu/~kay/courses/131/s04readings.html	375.473513
4	https://www.ics.uci.edu/~eppstein/bibs/eppstein.html	337.381997
5	https://www.ics.uci.edu/~eppstein/pubs/geom-all.html	263.919788

Screenshot

```
Query: ACM
1. https://www.ics.uci.edu/~eppstein/pubs/pubs.ff
    Score: 421.727496
2. https://www.ics.uci.edu/~eppstein/pubs/all.html
    Score: 389.077625
3. https://www.ics.uci.edu/~kay/courses/131/s04readings.html
    Score: 375.473513
4. https://www.ics.uci.edu/~eppstein/bibs/eppstein.html
    Score: 337.381997
5. https://www.ics.uci.edu/~eppstein/pubs/geom-all.html
    Score: 263.919788
```

Query 4

Query 4 is “master of software engineering”.

Results

Rank	URL	Score
1	https://www.ics.uci.edu/~kay/courses/h22/hw/DVD.txt	1898.100854
2	https://www.ics.uci.edu/~kay/courses/h22/hw/DVD-random.txt	1898.100854
3	https://www.ics.uci.edu/~eppstein/bibs/kpath.bib	822.442486
4	https://www.ics.uci.edu/~gbowker/classification/	719.055952
5	https://www.ics.uci.edu/~dan/class/267P/datasets/calgary/news	488.366396

Like with Query 2, this Query 4 output has a clear instance of two non-duplicate URLs with duplicate content both appearing on the ranking together. If these sorts of duplicates should be avoided, an alternative 5th row is the following:

6	https://www.ics.uci.edu/~pattis/quotations.html	417.178396

Screenshot

```
Query: master of software engineering
1. https://www.ics.uci.edu/~kay/courses/h22/hw/DVD.txt
    Score: 1898.100854
2. https://www.ics.uci.edu/~kay/courses/h22/hw/DVD-random.txt
    Score: 1898.100854
3. https://www.ics.uci.edu/~eppstein/bibs/kpath.bib
    Score: 822.442486
4. https://www.ics.uci.edu/~gbowker/classification/
    Score: 719.055952
5. https://www.ics.uci.edu/~dan/class/267P/datasets/calgary/news
    Score: 488.366396
```

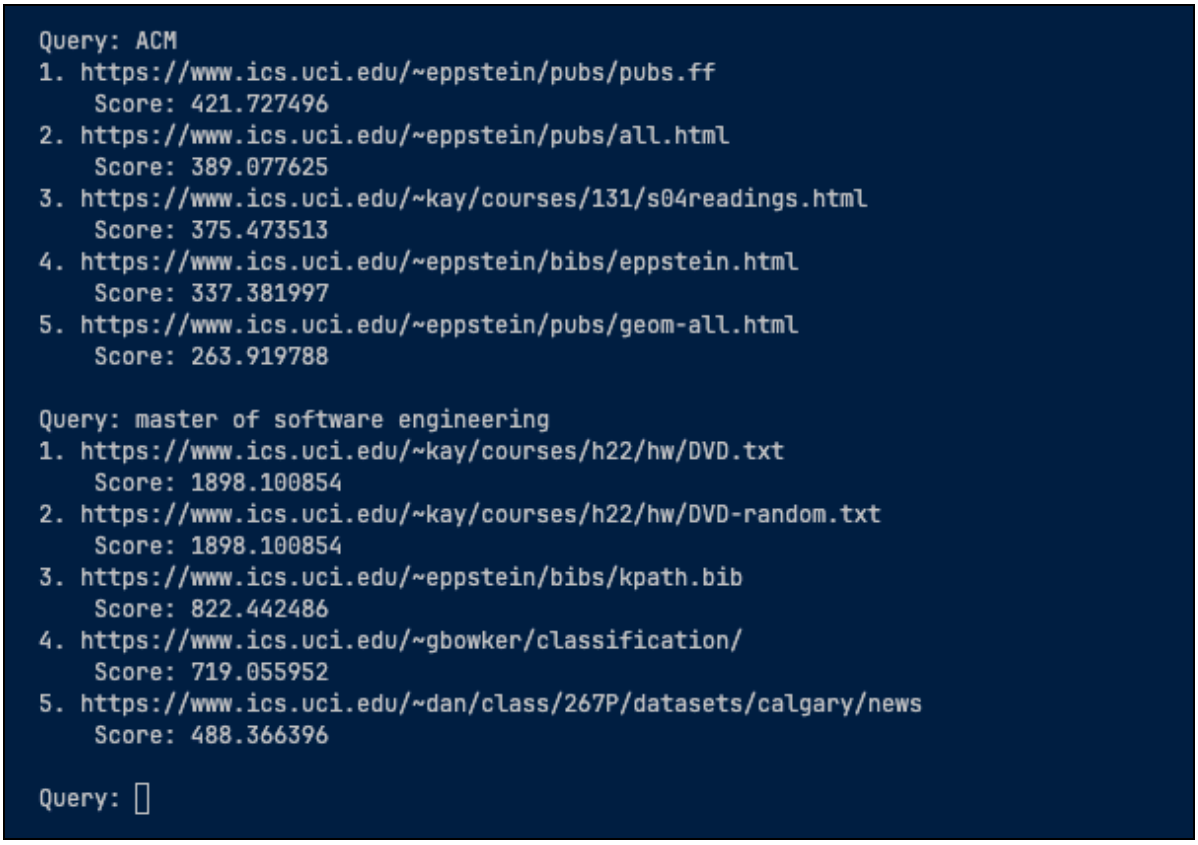
Full Search Interface Screenshot

The interface is being run in a VSCode terminal.

```
(.venv)                          a3_search_engine % python3 -m scripts.search
Loaded 55,393 documents.
Loaded 1,082,386 indexed terms.
Enter a search query. Use an empty query to quit.

Query: cristina lopes
1. https://www.ics.uci.edu/~hsajnani/
    Score: 231.226587
2. https://www.ics.uci.edu/community/news/notes/notes_2010.php
    Score: 116.009716
3. https://www.ics.uci.edu/~sjavanma/
    Score: 102.055391
4. https://www.informatics.uci.edu/explore/faculty-profiles/cristina-lopes/
    Score: 95.280054
5. http://mondego.ics.uci.edu/
    Score: 79.732807

Query: machine learning
1. http://mondego.ics.uci.edu/datasets/maven-contents.txt
    Score: 1129.465597
2. https://cml.ics.uci.edu/category/aiml/page/2/
    Score: 533.433817
3. https://cml.ics.uci.edu/category/aiml/
    Score: 487.634050
4. https://www.ics.uci.edu/faculty/index.php
    Score: 423.289227
5. https://www.ics.uci.edu/faculty/
    Score: 423.289227
```



Other Information

See the `README.md` file for a broader overview of the search implementation.